A Digital-Twin Multi-Agent Framework for Enterprise Cognitive Networking: ECNetBench and Leakage-Safe Multi-Seed Evaluation

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***Abstract*—Enterprise cognitive networking—combining digital twins, predictive analytics, explainable root-cause analysis (RCA), and closed-loop remediation support—lacks a shared, leakage-safe benchmark that couples realistic multi-layer enterprise configuration with multi-modal telemetry and supervised task labels. We present *ECNetBench*, a synthetic enterprise benchmark grounded in AOS-CX-style operational assumptions, together with an *Enterprise Cognitive Network* (ECN-v3) comprising Perception, Anomaly, Prediction, RCA, Impact, and Healing agents. The final T1 detector uses leakage-safe temporal feature enrichment with *anchored* telemetry-first fusion (`ECNFusionModel`); nesting-safe stacking is evaluated as an ablation and does not improve mean T1 AUPRC. RCA uses random-forest classification with TreeSHAP explanations. Evaluation follows a temporal $70/15/15$ protocol on six frozen instances (v1.1.0-INST and seeds 101–505). Across seeds, the final ECN-v3 detector attains mean T1 AUPRC of 0.11522078115707056, exceeding random forest (0.07583814878524626) and the prior v2 configuration (0.05771284608153401). The recommended T2 head is telemetry logistic regression (0.038043099063649714). We do not claim universal superiority on every paired test (n=6). Instead, we contribute (i) a reproducible multi-seed benchmark with leakage rules, (ii) a complete detection–prediction–RCA–impact–healing decision-support pipeline with an evidence-selected fusion head, and (iii) honest multi-seed confidence intervals, calibration, and paired significance tests.**

***Index Terms*—Enterprise networking, digital twin, multi-agent systems, anomaly detection, failure prediction, root-cause analysis, AIOps, benchmark, reproducibility.**

I. Introduction

Modern enterprise campuses and data-center fabrics are operated under continuous change: VLAN and ACL updates, EVPN/VXLAN overlays, QoS policy edits, and firmware-driven control-plane behavior. Cognitive NetOps and AIOps promise earlier anomaly detection, failure-horizon prediction, explainable RCA, service-impact estimation, and remediation recommendations [1]–[3]. In practice, progress is slowed by fragmented public resources: topology archives without multi-modal telemetry [4], security flow corpora that omit enterprise L2/L3 configuration semantics [5], and telemetry/RCA artifacts that do not jointly expose recovery and impact labels under a temporal leakage protocol.

This paper addresses that gap with two coupled contributions. First, **ECNetBench** is a synthetic enterprise cognitive-networking benchmark whose schema, failure taxonomy, and label definitions are designed for leakage-safe supervised learning under AOS-CX-style operational assumptions (inventory, VLAN/ACL/QoS, routing, counters, syslog/alerts, NAE-like analytics, and recovery/impact fields). Second, we implement and evaluate an **Enterprise Cognitive Network (ECN-v3)**: a Digital Twin graph plus a multi-agent stack (Perception, Anomaly, Prediction, RCA, Impact, Healing) with leakage-safe feature enrichment, *anchored* telemetry-first fusion (`ECNFusionModel`), and TreeSHAP-explained RCA.

a) Novelty statement.: To our knowledge, this is the first end-to-end twin + multi-agent detection–prediction–RCA–impact–healing decision-support pipeline evaluated on a frozen multi-seed enterprise benchmark under an explicit temporal $70/15/15$ leakage protocol with paired Wilcoxon tests and Cliff's δ across six independent instances.

b) Contributions.:

1) **ECNetBench design:** relational+graph schema, failure taxonomy, causal incident modeling, leakage-safe labels, realism validation, and six frozen evaluable instances (v1.1.0-INST; seeds 101, 202, 303, 404, 505).
2) **ECN-v3 framework:** Digital Twin construction; leakage-safe temporal enrichment; anchored fusion selected over nesting-safe stacking by architecture isolation; RF+TreeSHAP RCA.
3) **Rigorous multi-seed evaluation:** baselines, fusion ablations, calibration, confidence intervals, significance tests, and computational cost—with honest reporting of non-significant paired tests where n=6 is underpowered.
4) **Reproducible package:** relative-path evaluation harness, checksums, traceability artifacts, and a synchronized Overleaf project under `paper/overleaf/`.

c) Headline results (exact verified means, n=6).: Final ECN-v3 T1 AUPRC = 0.11522078115707056 versus strongest baseline random forest = 0.07583814878524626 and versus the prior v2 configuration = 0.05771284608153401. Nesting-safe stacking on the same enriched features is a *negative* ablation (0.0997458645152051). The recommended T2 head is telemetry logistic regression (0.038043099063649714). Twin contribution under the final T1 head is near zero ($\approx +0.00035$).

The overall architecture is introduced in Fig. 2 (Section IV). Section II situates related work; Sections III–IV detail the benchmark and framework; Sections V–VI present protocol and results; Sections VII–IX discuss implications, limitations, and conclusions.

II. Related Work

A. Network digital twins and autonomous networks

Network digital twins (NDTs) aim to provide data-driven, near-real-time models for what-if analysis and closed-loop control [3], [6]. Standards bodies and industry white papers discuss NDTs as enablers of autonomous networks and safe change validation [7], [8]. Our twin focuses on enterprise campus/DC inventory and telemetry graphs rather than packet-level WAN simulation, and is evaluated as a feature provider inside a multi-agent NetOps pipeline.

B. AIOps, anomaly detection, and RCA

Surveys of AIOps and network anomaly detection highlight class imbalance, concept drift, and the need for operationally meaningful metrics beyond accuracy [1], [2], [9]. Classical detectors (Isolation Forest [10], EWMA) and strong tabular learners remain competitive when features are informative. RCA research spans log mining, causal graphs, and Clos-fabric diagnostics [2]. We evaluate explainable category-level RCA with ablations that remove twin or syslog cues, and we treat healing as *decision support* rather than live switch actuation.

C. Graph learning for networks

Graph neural networks and message-passing proxies have been proposed for topology-aware prediction [11], [12]. We include a GraphSAGE-style proxy baseline for fairness, while the proposed ECN uses an explicit Digital Twin graph with neighbor aggregates rather than claiming end-to-end GNN superiority.

D. Benchmarks and datasets

Public resources include Topology Zoo [4], SNDlib [13], and security datasets such as UNSW-NB15 [5]. Enterprise cognitive NetOps additionally needs configuration semantics, multi-modal telemetry, service impact, and recovery labels under temporal freeze rules. ECNetBench is synthetic by design; we therefore emphasize leakage audits, seed robustness, and explicit threats to external validity (Section VIII).

E. Intent-based and multi-agent network management

Intent-based networking and multi-agent orchestration are long-standing themes in automated management [14], [15]. ECN-v3 instantiates specialized agents with an orchestrator; fusion is constrained (anchored telemetry weight ≥ 0.5) after architecture isolation showed nesting-safe stacking does not improve mean T1 AUPRC on ECNetBench. RCA explanations use TreeSHAP on a random-forest category model.

III. ECNetBench: Benchmark Design

A. Design goals

ECNetBench targets enterprise cognitive networking research with five goals: (G1) AOS-CX-style multi-layer configuration realism; (G2) multi-modal telemetry (counters, resources, syslog/alerts, flow aggregates); (G3) causal incident chains linking failure, detection, recovery, and impact; (G4) leakage-safe supervised labels; (G5) multi-seed robustness under frozen instances. Table I contrasts ECNetBench with common public resource classes for cognitive NetOps research.

TABLE I
Qualitative comparison of ECNetBench against common public resource classes for enterprise cognitive NetOps research.

Resource class	Config.	Telem.	RCA/heal	Leak.-safe multi-seed
Topology archives	Partial	No	No	Rare
Security flow sets	Low	Flows	Limited	Varies
Clos/telemetry RCA	Fabric	Yes	RCA	Varies
ECNetBench	**Yes**	**Yes**	**Yes**	**Yes**

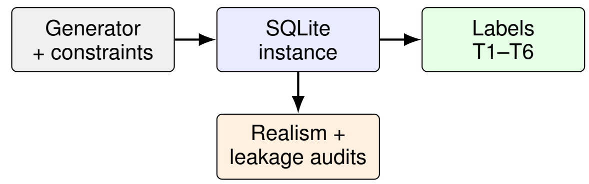

Fig. 1. ECNetBench generation and validation workflow: constrained synthesis produces a frozen SQLite instance that is audited for realism/leakage and exported to supervised task labels.

B. Enterprise assumptions and topology

Instances model a compact enterprise fabric (19 devices, 31 links in the evaluated family) spanning campus/DC roles with VLAN membership, ACL/QoS bindings, routing (e.g., BGP/OSPF-related tables), LAG/VSX-like redundancy concepts, and overlay-related objects where applicable. Topology snapshots and typed graph edges support Digital Twin construction. Generation constraints and schema documents accompany the release.

C. Telemetry and causal incident modeling

Synthetic generators emit interface counters, device resources, environmental/power samples, syslog/alert streams, and service KPIs. Failure incidents are drawn from a taxonomy covering link, device, control-plane, policy, and service-impacting families. Each incident carries detection, recovery-action, and impact fields so that labels for anomaly windows, failure horizons, RCA categories, impact, and healing actions remain consistent with a causal narrative. Fig. 1 summarizes the generation and validation workflow from constrained synthesis to audited SQLite instances and supervised labels.

D. Leakage-safe labels and splits

Labels are aligned to time bins. Features at decision time t_0 use only information available at or before t_0 (historical shifts; no future counters). Evaluation uses a temporal $70/15/15$ train/validation/test split frozen per instance. Forbidden fields (raw future labels, oracle incident IDs in feature matrices) are audited in the companion leakage reports.

TABLE II
FROZEN EVALUATION INSTANCES USED IN THIS STUDY (AUTHORITATIVE SQLITE STORES).

Instance folder	Role	Devices / links
`v1` (v1.1.0-INST)	Primary frozen instance	19 / 31
`v1.1-seed101`	Independent seed	19 / 31
`v1.1-seed202`	Independent seed	19 / 31
`v1.1-seed303`	Independent seed	19 / 31
`v1.1-seed404`	Independent seed	19 / 31
`v1.1-seed505`	Independent seed	19 / 31

TABLE III
SUMMARY OF ECNETBENCH RELATIONAL SCHEMA FAMILIES.

Family	Representative contents
Inventory/topology	Devices, interfaces, topology snapshots, graph nodes/edges
Configuration	VLAN/ACL/QoS, routing processes, configuration snapshots/diffs
Telemetry	Interface counters, resources, syslog/alerts, flow aggregates
Services	Applications, services, SLA and impact bindings
Incidents/labels	Failure incidents, recovery actions, supervised label tables

E. Instances and seed robustness

We evaluate six frozen SQLite instances: `v1.1.0-INST` and independent seeds 101, 202, 303, 404, and 505, as listed in Table II. SHA-256 digests are published in `benchmark/INSTANCE_CHECKSUMS.json`. Seed-robustness and publication-validation reports accompany the package. SQLite is authoritative for evaluation; CSV/Parquet exports are optional derivatives. Schema families are summarized in Table III.

IV. ENTERPRISE COGNITIVE NETWORK FRAMEWORK

A. Digital Twin

The Digital Twin materializes inventory and topology as a typed graph synchronized with telemetry windows. Twin-derived features include residual contrasts (e.g., device resource versus neighbor summaries), structural aggregates, and interface error/discard/carrier deltas constructed with causal shifts. Twin features are optional inputs to agents and are ablated in `no_twin` / `twin_only` settings.

B. Leakage-safe feature enrichment (ECN-v3)

Beyond legacy aggregates, ECN-v3 adds causal rolling/EMA statistics, second differences, error-burst accumulators, and neighbor-instability proxies. All rolling and expanding operators apply `shift(1)` so that the current bin is excluded; labels join features at $\texttt{feat\_bin} = t_{\text{start}} - 30\,\text{min}$. Isolation studies attribute the T1 gain primarily to this enrichment rather than to fusion complexity.

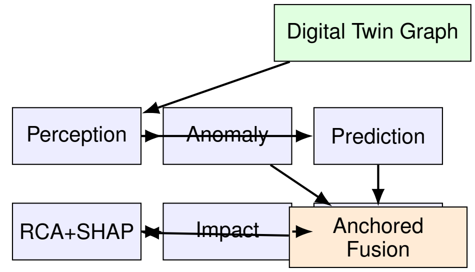

Fig. 2. ECN-v3 multi-agent workflow: Perception feeds Anomaly/Prediction specialists into *anchored* orchestrated fusion, followed by TreeSHAP RCA, impact estimation, and healing decision support.

C. Multi-agent architecture

Fig. 2 depicts the agent workflow. Perception builds leakage-safe feature matrices from telemetry and twin views. Anomaly and Prediction score T1 anomaly windows and T2 failure-horizon risk, respectively. RCA predicts failure category with TreeSHAP explanations (T3); Impact estimates service-impact labels (T4); and Healing recommends remediation classes as *decision support* (TR-AUTO). Actions are not executed on physical switches in this study.

a) Final fusion head.: Architecture selection under the frozen six-seed protocol compares nesting-safe stacking against anchored fusion on the same enriched features. Anchored fusion (`ECNFusionModel`) constrains the telemetry specialist weight to be at least 0.5 (or selects telem-only), yielding higher mean T1 AUPRC with a simpler operator-facing interpretation. Stacking (`ECNStackFusionModel`) is retained as an ablation/negative result. The recommended T2 head is telemetry logistic regression under ultra-rare positives.

D. Class imbalance and calibration

Positive rates are low ($\sim$1% for T1; lower for T2). Training uses imbalance-aware weighting where applicable (`scale_pos_weight`); Isolation Forest scores are mapped via a train empirical CDF. Thresholds for F1/precision/recall are tuned on validation only and are *not* used for AUPRC. Primary ranking metrics are AUPRC and ROC-AUC; Brier score reports calibration. Optional Beta/Platt calibration may be applied for probability display without changing AUPRC ranking.

V. EVALUATION PROTOCOL

A. Tasks

We report T1 anomaly detection, T2 failure-horizon prediction, T3 RCA (macro-F1 with TreeSHAP), T4 impact, and TR-AUTO healing decision support. Supplementary tasks (degradation, config risk) are produced by the harness and archived in `results/`.

B. Baselines and fairness

Baselines include logistic regression, random forest, LightGBM, Isolation Forest, EWMA, threshold rules, majority class, an MLP sequence proxy, and a GraphSAGE-style GNN proxy. Baselines train on telemetry-only feature matrices; the proposed T1 model uses twin+telemetry with **anchored** fusion on leakage-safe enriched features. Shared temporal enrichment is applied consistently so comparisons are not confounded by unequal feature engineering for baselines versus proposed specialists.

C. Metrics and statistics

Primary metric: average precision (AUPRC). Secondary: ROC-AUC, precision, recall, F1 (validation-tuned threshold), Brier score. Aggregation: mean and 95% CI across six seeds. Significance: paired Wilcoxon signed-rank tests and Cliff's δ of the final T1 head versus each baseline and versus the stacking ablation on per-seed AUPRC.

D. Ablations

Primary architecture ablations contrast (i) v2 legacy features + anchored fusion, (ii) v3 enriched features + stacking, and (iii) v3 enriched features + anchored fusion (final). Module deltas report feature-enrichment gain versus v2, twin contribution under the final head, and stacking versus anchored on identical features.

E. Computational cost

Full-suite wall-clock time averages approximately 41 s per seed (95% CI $[37.83, 43.72]$). Mean T1 training time for the anchored head is 2.21 s, comparable to stacking (2.28 s) and larger than RF telem_only (0.59 s).

VI. Results

All numerical claims below are taken from verified artifacts `results/manuscript_ready_numbers.json`, `results/v3_gated/t1_architecture_selection.json`, and `results/aggregate_v3.json` (baselines/T2/RCA), without modifying frozen instances.

A. T1 anomaly and T2 failure prediction

Table IV summarizes multi-seed mean ranking metrics. On T1, the final ECN-v3 anchored detector achieves mean AUPRC 0.11522078115707056 (95% CI $[0.0609, 0.1695]$), exceeding random forest (0.07583814878524626) and the v2 legacy configuration (0.05771284608153401). Mean T1 ROC-AUC is 0.7533587867665612. Fig. 3 visualizes the same AUPRC means with 95% confidence intervals for T1 and T2, highlighting the T1 lift of the final head and the T2 telem-logistic recommendation.

On T2, the recommended head is telemetry logistic regression with mean AUPRC 0.038043099063649714, higher than RF (0.017610244849363122). We do not claim that multi-agent fusion improves T2 under ultra-rare positives.

Paired Wilcoxon tests and Cliff's δ (Table V) show that the T1 mean advantage versus random forest is not significant at α=0.05 (p=0.688, δ=+0.389), consistent with n=6 underpowering. Versus logistic and several weaker baselines, paired p-values reach 0.031. Stacking versus anchored is not significant (p=0.375).

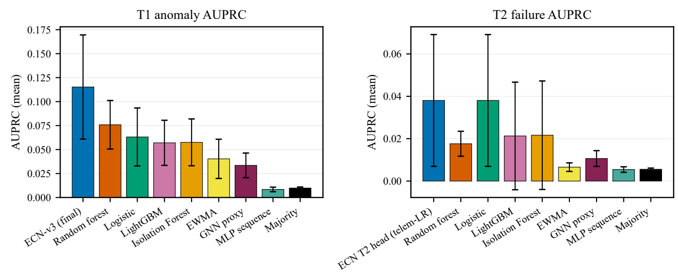

Fig. 3. Multi-seed mean AUPRC with 95% confidence intervals. Left: T1 anomaly detection, where ECN-v3 denotes the final anchored head on enriched features. Right: T2 failure-horizon ranking; the ECN T2 column is the recommended telemetry logistic head.

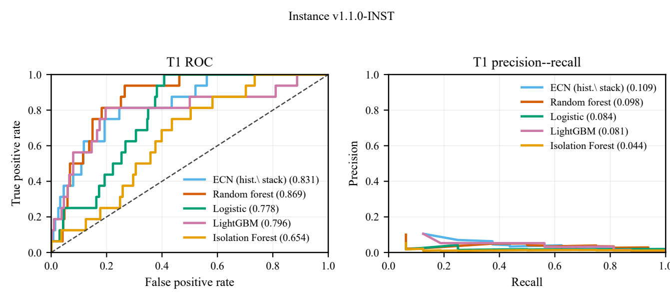

Fig. 4. T1 ROC (left) and precision–recall (right) curves on frozen instance v1.1.0-INST for representative detectors. Legend values report ROC-AUC and average precision on this instance.

Representative ROC and precision–recall curves on the primary frozen instance appear in Fig. 4. Curve geometry remains consistent with rare-positive ranking: ROC discrimination is moderate, while precision–recall is challenging for all methods. (Stored per-seed curves include a historical stacking proposed trace for continuity; mean claims use the final anchored artifact.)

B. Architecture selection and fusion ablation

Fig. 5 and Table VI isolate the sources of the T1 improvement. Leakage-safe feature enrichment accounts for the dominant gain; replacing anchored fusion with stacking on the same enriched features *reduces* mean AUPRC from 0.1152 to 0.0997. Accordingly, stacking is reported as an ablation/negative result rather than as the final T1 architecture.

Table VII and Fig. 6 summarize verified contribution deltas. Feature enrichment versus v2 contributes $+0.0575$ mean AUPRC. Twin contribution under the final head is near zero ($+0.0003$). Stacking relative to anchored contributes -0.0155.

C. Calibration and operating points

Calibration quality is summarized in Table VIII. Final T1 raw Brier is 0.1509, worse than RF (0.0227), motivating optional Beta calibration for probability displays without altering AUPRC ranking. Fig. 7 shows a reliability diagram on v1.1.0-INST for stored ECN scores.

D. RCA, healing decision support, and cost

Table IX reports RCA macro-F1, healing decision-support scores, and cost. Proposed T3 RCA macro-F1 mean is

TABLE IV
MULTI-SEED MEAN DETECTION METRICS ON T1 (ANOMALY) AND T2 (FAILURE HORIZON). ECN-V3 USES LEAKAGE-SAFE ENRICHED FEATURES WITH **ANCHORED** FUSION ON T1; THE RECOMMENDED T2 HEAD IS TELEMETRY LOGISTIC REGRESSION. AUPRC ENTRIES REPORT MEANS OVER SIX FROZEN INSTANCES WITH 95% CONFIDENCE INTERVALS.

Method	T1 AUPRC	T1 ROC-AUC	T2 AUPRC	T2 ROC-AUC
ECN-v3 (final)†	**0.1152** [0.0609, 0.1695]	**0.7534**	0.0380 [−0.0027, 0.0788]	0.6891
Stacking ablation	0.0997 [0.0738, 0.1257]	0.7803	—	—
v2 legacy+anchored	0.0577 [0.0214, 0.0940]	0.6831	—	—
Random forest	0.0758 [0.0505, 0.1011]	0.7661	0.0176 [0.0099, 0.0253]	0.7081
Logistic	0.0631 [0.0329, 0.0934]	0.7260	**0.0380** [−0.0027, 0.0788]	0.6891
LightGBM	0.0570 [0.0335, 0.0805]	0.6798	0.0213 [−0.0041, 0.0467]	0.5483
Isolation Forest	0.0575 [0.0331, 0.0819]	0.6716	0.0216 [−0.0040, 0.0472]	0.6459
EWMA	0.0403 [0.0198, 0.0607]	0.5484	0.0066 [0.0045, 0.0086]	0.4482
GNN proxy	0.0335 [0.0167, 0.0504]	0.6846	0.0106 [0.0069, 0.0144]	0.5965
MLP sequence	0.0084 [0.0053, 0.0115]	0.3617	0.0054 [0.0041, 0.0067]	0.3795
Majority	0.0098 [0.0083, 0.0113]	0.5000	0.0056 [0.0050, 0.0061]	0.5000

†T1: `ECNFusionModel` on v3 features; T2 column reports the recommended telem-LR head (identical to Logistic on T2).
Stacking and v2 rows are historical/ablation references, not the final proposed detector.

TABLE V
PAIRED WILCOXON SIGNED-RANK TESTS AND CLIFF'S δ FOR T1 AUPRC OF THE FINAL ECN-V3 ANCHORED MODEL VERSUS BASELINES AND THE STACKING ABLATION (n=6). POSITIVE δ FAVORS ECN-V3.

Comparator	p	Cliff δ	ECN-v3 mean	Comparator mean
Random forest	0.688	+0.389	0.1152	0.0758
Logistic	0.031	+0.556	0.1152	0.0631
LightGBM	0.062	+0.611	0.1152	0.0570
Isolation Forest	0.031	+0.667	0.1152	0.0575
EWMA	0.062	+0.778	0.1152	0.0403
GNN proxy	0.062	+0.778	0.1152	0.0335
MLP sequence	0.031	+1.000	0.1152	0.0084
Majority	0.031	+1.000	0.1152	0.0098
Stacking ablation	0.375	+0.111	0.1152	0.0997

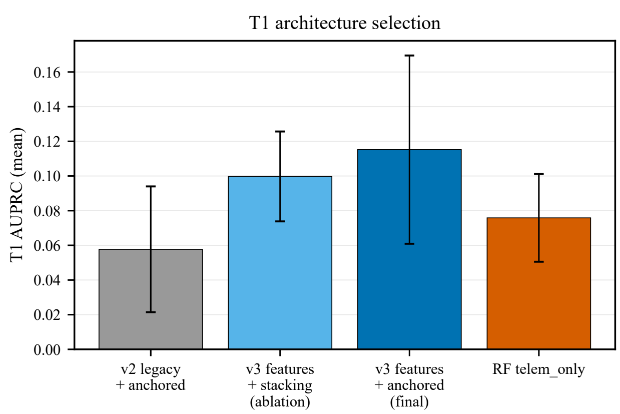

Fig. 5. T1 architecture selection: v2 legacy+anchored, v3+stacking (ablation), v3+anchored (final), and RF telem_only. Error bars denote 95% confidence intervals over six seeds.

0.25396825396825395 using RF with TreeSHAP explanations. Healing with RCA category features reaches macro-F1 1.0 on this synthetic setup—an upper bound that is *not* interpreted as operational perfection; the honest `no_rca_cat` setting yields 0.19027777777777777. Fig. 8 illustrates a representative T3 confusion matrix on the primary instance.

TABLE VI
T1 ARCHITECTURE AND FUSION ABLATION UNDER THE FROZEN SIX-SEED PROTOCOL. STACKING IS A NEGATIVE RESULT RELATIVE TO ANCHORED FUSION ON ENRICHED FEATURES.

Configuration	Mean AUPRC	95% CI
v2 legacy features + anchored	0.0577	[0.0214, 0.0940]
v3 enriched features + stacking (ablation)	0.0997	[0.0738, 0.1257]
v3 enriched + anchored (final)	**0.1152**	[0.0609, 0.1695]
RF telem_only (strongest baseline)	0.0758	[0.0505, 0.1011]

TABLE VII
VERIFIED T1 CONTRIBUTION DELTAS UNDER THE FINAL PROTOCOL. FEATURE ENRICHMENT IS THE DOMINANT GAIN; TWIN CONTRIBUTION IS NEAR ZERO; STACKING REDUCES MEAN AUPRC RELATIVE TO ANCHORED FUSION.

Contribution	Δ T1 AUPRC
Feature enrichment (final − v2)	+0.0575
Twin (full − no_twin) under final head	+0.0003
Stacking − anchored (on v3 features)	−0.0155

E. Why simplicity remains preferable

At ∼1% positives, enriched telemetry recovers most ranking signal. Anchored fusion is simpler than stacking, improves

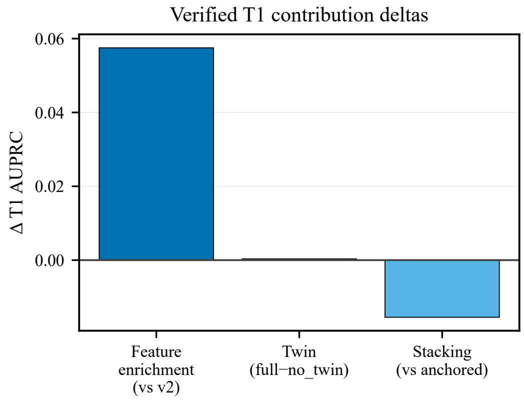

Fig. 6. Verified T1 AUPRC deltas: feature enrichment versus v2, twin contribution under the final head, and stacking versus anchored fusion.

TABLE VIII

Calibration quality (Brier score; lower is better). Final ECN-v3 T1 Brier is reported from the architecture-selection study; baseline Brier values are from the six-seed aggregate. Optional Beta calibration is recommended for probability display and does not change AUPRC ranking.

Task	Method	Mean	Notes
T1	ECN-v3 final (raw)	0.1509	Optional Beta for display
T1	Random forest	0.0227	Strong tabular calibration
T1	Logistic	0.2328	
T2	Telem logistic (T2 head)	0.2577	Recommended T2 head
T2	Random forest	0.0124	

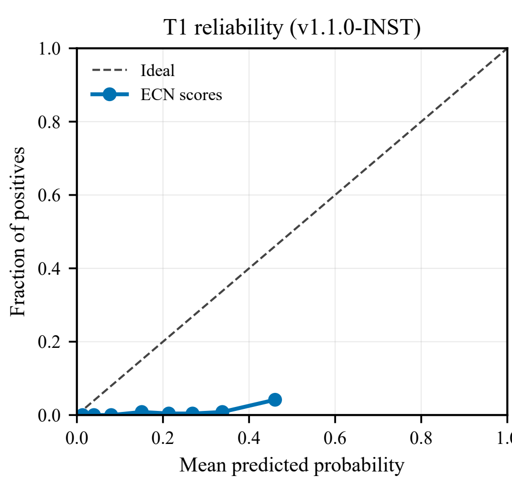

Fig. 7. T1 reliability diagram on instance v1.1.0-INST. The dashed line denotes perfect calibration. Optional Beta calibration is recommended for operator-facing probabilities.

mean T1 AUPRC on the same features, and remains competitive with RF in mean while acknowledging non-significant paired tests versus RF. With n=6 seeds, intervals are wide; claims emphasize reproducible mean improvements and architecture honesty rather than universal paired dominance.

TABLE IX

RCA accuracy, healing decision-support scores, and evaluation wall-clock cost (six seeds). RCA uses RF with TreeSHAP explanations. The full healing setting includes RCA category features; no_rca_cat is the honest telemetry-oriented baseline.

Metric	Setting	Mean [95% CI]
T3 RCA macro-F1	RF + TreeSHAP	0.2540 [0.0901, 0.4179]
Healing macro-F1	full (w/ RCA cat.)	1.0000 [1.0000, 1.0000]
Healing macro-F1	no_rca_cat	0.1903 [0.0989, 0.2816]
T1 train time (s)	anchored head	2.21
Eval. wall time (s/seed)	full suite	∼41 [37.83, 43.72]

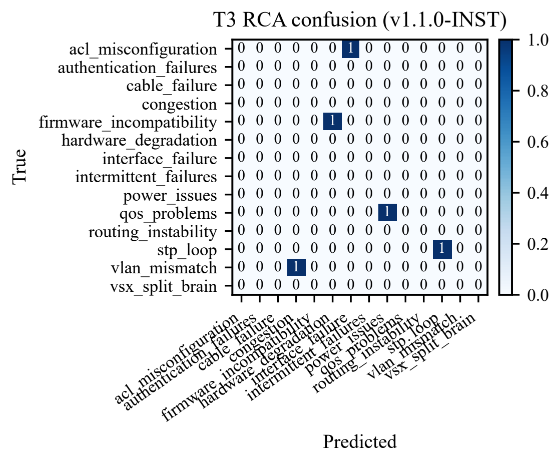

Fig. 8. Representative T3 RCA confusion matrix for the RF+TreeSHAP RCA head on instance v1.1.0-INST.

VII. Discussion

A. What to claim

The defensible claims are architectural and methodological: (1) ECNetBench provides a leakage-aware multi-task enterprise benchmark with multi-seed frozen instances; (2) ECN-v3 closes the loop from detection to healing *decision support* with TreeSHAP RCA; (3) leakage-safe temporal enrichment is the primary driver of the T1 AUPRC lift from 0.0577 (v2) to 0.1152 (final); (4) anchored fusion is preferred over stacking on this benchmark (higher mean AUPRC, lower complexity); (5) T2 remains best served by a telem logistic head under extreme rarity.

B. What not to claim

We do not claim statistically significant superiority over random forest on T1 (p=0.688), significant superiority of anchored over stacking (p=0.375), live deployment on physical AOS-CX switches, or that multi-agent fusion alone beats strong tabular detectors without feature enrichment. Impact AUPRC of 1.0 for some models on this synthetic label construction is reported for completeness but interpreted cautiously as potential label/feature collinearity rather than solved service-impact prediction in the wild.

C. Practical NetOps implications

Operators can use ECN as a sandbox for ranking alerts, proposing RCA categories with SHAP attributions, and sug-

TABLE X
THREATS TO VALIDITY AND CORRESPONDING MITIGATIONS.

Threat	Mitigation / residual risk
Synthetic data	Realism audits; still not a substitute for production traces.
External validity	AOS-CX-style assumptions; other vendors/fabrics may differ.
Label leakage	Temporal freeze and feature audits; healing full vs. `no_rca_cat`.
Small n seeds	Six seeds with confidence intervals; tests remain underpowered.
Topology scale	19 devices / 31 links; scalability to large fabrics unproven.
Healing semantics	Decision support only; no live actuation evidence.

gesting remediation classes before human approval. Prefer the anchored head for T1 triage; prefer telem logistic scores for T2 horizon ranking; apply optional Beta calibration when displaying probabilities.

VIII. LIMITATIONS AND THREATS TO VALIDITY

Table X summarizes the principal threats to validity. Construct validity risks include synthetic incident generators that may over-structure causal chains. Internal validity is strengthened by frozen instances, checksums, and relative-path reproduction. Conclusion validity is limited by n=6 and wide confidence intervals. External validity requires future evaluation on anonymized production telemetry.

IX. CONCLUSION

We presented ECNetBench and an Enterprise Cognitive Network (ECN-v3) digital-twin multi-agent framework evaluated under leakage-safe multi-seed protocols. Exact verified means show final T1 AUPRC 0.11522078115707056 for leakage-safe enriched features with anchored fusion, exceeding random forest 0.07583814878524626 and the v2 configuration 0.05771284608153401. Stacking on the same enriched features is a negative ablation (0.09974586451520 51). The recommended T2 head is telemetry logistic regression (0.038043099063649714). RCA uses RF with TreeSHAP. The primary scientific value is a reproducible benchmark plus a complete cognitive NetOps pipeline with evidence-selected fusion and honest statistics—not a universal detector champion. Future work includes larger topologies, production-trace calibration, and human-in-the-loop healing studies.

APPENDIX A
SUPPLEMENTARY NOTES

A. Reproduction commands

```
python -m venv .venv
pip install -r requirements.txt
python scripts/download_instances.py
python scripts/verify_instances.py
pytest -q
python evaluation/trace_t1_gain.py
python evaluation/select_t1_architecture.py
python evaluation/run_full_evaluation.py
python paper/overleaf/scripts/regenerate_pub_figur
pdflatex main.tex && bibtex main && pdflatex main.
```

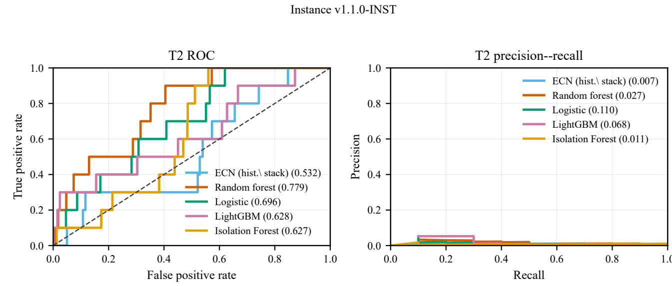

Fig. 9. T2 ROC (left) and precision–recall (right) curves on frozen instance v1.1.0-INST. The recommended T2 operating head is telemetry logistic regression.

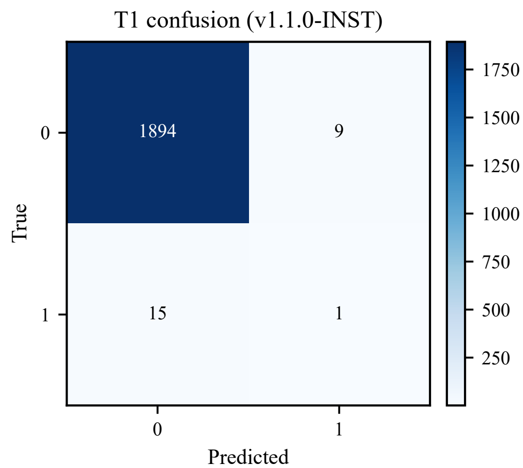

Fig. 10. T1 confusion matrix on instance v1.1.0-INST at a validation-tuned threshold (threshold unused for AUPRC).

B. Traceability

Final T1 numbers map to `results/manuscript_ready_numbers.json` and `results/v3_gated/t1_architecture_selection.json`. Baseline/T2/RCA aggregates map to `results/aggregate_v3.json` and `results/per_seed/*.json`. Additional reports: `reports/FINAL_ARCHITECTURE_SELECTION.md`, `reports/T1_IMPROVEMENT_TRACEABILITY.md`. The Overleaf project is synchronized under `paper/overleaf/`.

C. Additional figures

Per-seed ROC/PR, confusion matrices, and calibration plots for T1/T2/T3 are included under `figures/` as vector PDF and 600 dpi PNG pairs. T2 curves and the T1 confusion matrix for the primary instance are shown below; remaining seeds are archived for completeness.

DATA AVAILABILITY

Frozen ECNetBench SQLite instances and checksums are distributed with the companion repository https://

github.com/Audrey2023uh/Network-Multi-Agent (release tag `ecnetbench-v1.1.0-data`). Evaluation reads instances under relative path `benchmark/instances/`.

Code Availability

Source code for the generator, Digital Twin, agents, baselines, ablations, and evaluation harness is available at the same repository under an MIT license. The Overleaf manuscript source is synchronized under `paper/overleaf/`. Exact reproduction commands appear in the companion reproducibility notes.

Author Contributions

Conceptualization, methodology, software, validation, investigation, data curation, writing—original draft, and visualization were performed by the author(s). Formal analysis and statistical testing follow the published evaluation scripts.

Conflict of Interest

The authors declare no competing interests.

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